

UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-language models (VLMs) have progressed rapidly on visual question answering, document understanding, multimodal dialogue, and related evaluations, but aggregate gains on these benchmarks do not by themselves establish whether current systems retain the fine-grained visual knowledge needed to distinguish visually similar categories. We study this capability by converting four established recognition benchmarks into five-way multiple-choice VQA-style tasks, evaluating 15 recent VLMs, and relating their performance to eight general VLM benchmarks. The results show that fine-grained classification varies substantially among models with comparable general VQA scores, establishing it as a distinct vision-centric capability that standard benchmark aggregates measure only indirectly. We further find that most VLMs underperform their corresponding CLIP-style vision encoders under this protocol, suggesting that multimodal integration often fails to expose the full fine-grained recognition capacity already present in the visual backbone. Through 22 controlled LLaVA-style ablations, we identify the design and training factors that most consistently support this capability: stronger LLMs improve fine-grained and general benchmark scores in parallel, stronger vision encoders produce disproportionately large gains on fine-grained classification, and image-caption pretraining is most effective when the LLM is updated together with the connector. These findings clarify how fine-grained visual knowledge is acquired, transferred, and preserved under the evaluated VLM training recipe, and provide empirical guidance for designing multimodal systems with stronger vision-centric recognition behavior.

1 INTRODUCTION

Vision-language models (VLMs), which couple visual encoders with large language models (LLMs), now achieve strong results across open-ended visual question answering, document understanding, multimodal dialogue, and other image-conditioned reasoning tasks (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). As a consequence, general VQA benchmarks have become a primary instrument for measuring progress in multimodal modeling (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). Yet this benchmark-centered view leaves an essential vision-centric capability insufficiently resolved: whether contemporary VLMs can discriminate fine-grained visual categories, and which architectural and training decisions determine whether that knowledge survives multimodal alignment.

Fine-grained visual recognition—distinguishing categories that share substantial visual structure—requires attention to subtle discriminative cues while remaining robust to intra-class variation in pose, illumination, viewpoint, and background. Vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have therefore been evaluated extensively on fine-grained classification benchmarks and often attain strong recognition performance. Modern LLM-based VLMs, however, are typically assessed through VQA-oriented evaluations that entangle perception with language understanding, instruction following, and reasoning. Consequently, the extent to which these systems preserve, expose, and use fine-grained visual knowledge remains undercharacterized.

This capability is consequential because fine-grained perception frequently supplies the perceptual premise for downstream reasoning. Real-world visual inputs often follow long-tailed category distributions in which closely related classes imply materially different actions or conclusions. A model

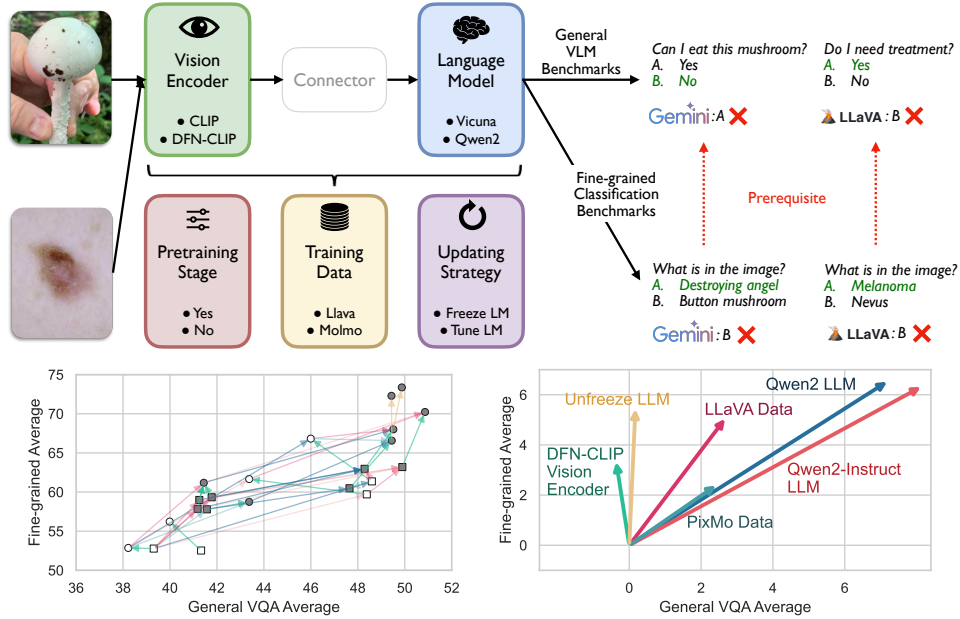


Figure 1: **Overview.** (Top) We evaluate fine-grained classification in vision-language models (VLMs) as a vision-centric capability that supports reliable higher-level visual understanding and reasoning. (Bottom) A suite of 22 controlled ablations over model components and training strategies (left) identifies the factors that most strongly affect fine-grained classification performance in the evaluated VLM pipeline (right).

that cannot identify a mushroom species, for example, cannot reliably use textual knowledge to infer whether the mushroom is poisonous or safe for consumption. Studying fine-grained recognition in VLMs therefore informs both evaluation and model design: it tests whether current systems retain the visual substrate required for reliable reasoning, and it reveals which integration choices preserve that substrate rather than obscuring it behind language-mediated benchmark performance.

This paper provides a systematic empirical account of fine-grained visual knowledge in contemporary VLMs and its relationship to standard general VLM evaluation. We evaluate 15 VLMs, including LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024), on four established fine-grained recognition benchmarks: ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014). We convert these datasets into five-way multiple-choice VQA-style tasks and compare the resulting scores with aggregated performance on eight general VLM benchmarks, including MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). This evaluation shows that models with similar general VQA performance can differ sharply in fine-grained classification accuracy, demonstrating that fine-grained recognition is a distinct axis of visual capability rather than a quantity reliably summarized by existing VLM benchmark aggregates. We also observe a persistent gap between most VLMs and their CLIP-based vision encoders, consistent with prior evidence that multimodal integration can leave visual-encoder classification knowledge only partially accessible to the language model (Zhang et al., 2024; Geigle et al., 2024).

We then perform a controlled ablation study to identify which components and training stages most directly shape fine-grained classification under a common LLaVA-style pipeline. The study varies the base language model, the vision encoder, the pretraining data source, the instruction-tuning data source, and the parameter-updating strategy. Across 22 ablations, the evidence isolates several design factors that govern fine-grained knowledge acquisition and preservation:

- **Language model (§4.2):** Stronger base language models improve fine-grained classification and general VLM benchmarks in a broadly uniform manner.
- **Vision encoder (§4.2):** Stronger vision encoders yield disproportionate gains on fine-grained classification, while their effect on general VLM benchmarks is more limited, particularly in two-stage training pipelines.

- **Pretraining stage (§4.3):** Large-scale image-caption pretraining substantially improves fine-grained classification and produces smaller gains on general VLM benchmarks.
- **Weight updating method (§4.3):** Fine-grained classification improves when pretraining updates both the connector between the vision encoder and LLM and the LLM itself, rather than updating only the connector.
- **Data quality (§4.3):** Within the evaluated settings, pretraining and instruction-tuning data quality has limited effect on fine-grained classification, especially when LLM weights remain frozen during pretraining.

Taken together, these results establish fine-grained visual perception as an important but under-measured component of VLM capability. They also show that improving this capability requires more than optimizing aggregate multimodal reasoning scores: the strength of the visual backbone, the presence and scale of image-caption pretraining, and the parameter-update strategy used during alignment all materially affect whether fine-grained visual knowledge remains usable. The resulting analysis offers a concrete empirical basis for developing more vision-centric VLMs in recognition settings that demand discrimination among visually similar and long-tailed categories.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) integrate visual inputs with language models, commonly through a LLaVA-style architecture (Liu et al., 2023) that combines a vision encoder, a language model, and an MLP connector that maps visual representations into the language-model input space. Recent systems have achieved strong performance on multimodal tasks including visual question answering (VQA), document understanding, and general visual reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). Progress in this area is frequently assessed with benchmarks such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). Although these evaluations are indispensable for measuring reasoning and language use conditioned on images, they provide only indirect evidence about core vision-centric abilities such as object recognition and fine-grained category discrimination. Our work targets this evaluation gap by measuring fine-grained classification in VLMs under a unified protocol and by showing that this capability is not adequately represented by standard VQA-oriented benchmark aggregates.

Fine-grained visual classification. Fine-grained classification is a long-standing computer vision problem in which models must distinguish visually similar subcategories within a broader semantic group, such as bird species (Wah et al., 2011), flower species (Nilsback & Zisserman, 2008), or pet breeds (Parkhi et al., 2012). These benchmarks have played a central role in developing and assessing visual encoders, including ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021). Despite this importance in vision research, recent VLM evaluation has largely shifted toward VQA benchmarks that jointly test perception, language understanding, and reasoning. Because many consequential visual settings involve long-tailed distributions and visually confusable classes—including medical diagnosis, food safety, and species identification—fine-grained classification remains a necessary probe of whether VLMs preserve precise visual knowledge beyond language-mediated reasoning performance.

Evaluation and ablation of VLMs for fine-grained classification. Recent studies have begun to examine VLMs on fine-grained classification tasks (Zhang et al., 2024; Geigle et al., 2024), but these analyses are often restricted to a smaller set of models or datasets. We extend this line of work by evaluating a broader collection of contemporary VLMs across multiple established fine-grained classification benchmarks using a common multiple-choice formulation. More importantly, we move beyond observational benchmarking by conducting controlled ablations that identify which model components and training decisions most strongly affect fine-grained classification performance.

Several related efforts study the VLM design space, including architectural choices (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and multimodal data composition (Gadre et al., 2023; Udandara et al., 2024; Deitke et al., 2024). These studies primarily emphasize general benchmarks that mix language understanding, reasoning, and coarse visual perception. In contrast, our analysis asks how base model selection, pretraining methodology, and the composition of pretraining and instruction-tuning data affect fine-grained classification specifically, thereby providing targeted evidence for optimizing VLMs as vision-centric recognition systems.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section establishes the empirical motivation for evaluating fine-grained classification in VLMs. We first define the recognition setting and benchmark conversion procedure, then describe the testing protocol for VLMs and vision-only encoders, and finally report the main observations from 15 recent VLMs.

3.1 MOTIVATION

Fine-grained classification is a fundamental visual recognition task in which a model must distinguish visually similar subcategories within a broader category, such as bird species or car models (Wei et al., 2021). Benchmarks such as ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008) have therefore served as standard testbeds for training and evaluating vision encoders. With large-scale supervised and self-supervised learning, encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have reached near- or superhuman performance on challenging fine-grained classification tasks.

Despite their central role in computer vision, these benchmarks are rarely foregrounded in VLM evaluation. Recent VLMs such as LLaVA (Liu et al., 2023) demonstrate strong image-conditioned instruction following, but they are typically evaluated through VQA benchmarks such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or through document- and text-oriented tasks such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). These benchmarks emphasize reasoning and language processing conditioned on images, while offering limited direct evidence about whether the model can discriminate fine-grained object categories.

Fine-grained recognition is not merely a peripheral visual skill for VLMs; it is often the perceptual condition for correct downstream reasoning. In everyday assistance, a model asked whether a mushroom is safe to eat must first identify the species before reasoning about toxicity; confusing a deadly destroying angel with an edible white button mushroom could have severe consequences. Analogously, medical decision support can depend on distinguishing visually similar diseases, and autonomous navigation can depend on separating signs with similar color and shape. These examples do not expand the empirical scope of this paper, but they make clear why the evaluated capability matters: robust VLMs should preserve precise visual knowledge when confronted with long-tailed and visually confusable categories.

3.2 BENCHMARKS

We evaluate fine-grained visual knowledge using four established object-recognition benchmarks that span distinct fine-grained domains: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

These datasets contain between 37 and 1,000 classes, making full-class multiple-choice presentation impractical for VLMs that are commonly prompted with compact answer sets. Following Geigle et al. (2024), we convert each dataset into a five-way multiple-choice benchmark. For each image, an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) selects the four hardest negative labels according to image-label cosine similarity; these negatives are then combined with the ground-truth label to form the answer choices. We use OpenCLIP for negative selection so that the benchmark is not constructed with the same visual backbone used by many evaluated VLMs, including OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), and DFN-CLIP (Fang et al., 2023), each of which is trained on data different from the OpenCLIP selector. The resulting protocol evaluates VLMs in a native multiple-choice format while preserving much of the original classification difficulty: CLIP ViT-L/14-336px (Radford et al., 2021) obtains 75.2% accuracy on the original 102-class Flowers-102 benchmark and 78.3% accuracy on our five-way version.

3.3 TESTING SETUP

We evaluate 15 recent VLMs in the 7B–13B parameter range on the converted five-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abdin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b).

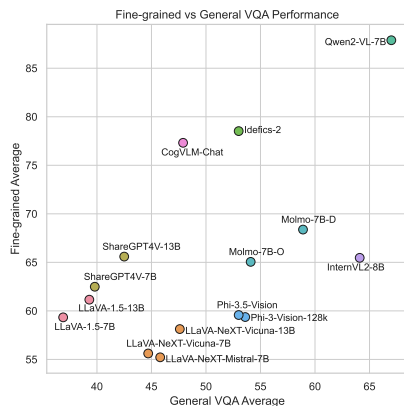


Figure 2: **Fine-grained classification versus general VQA performance across VLMs.** Models with similar aggregate VQA scores can differ substantially in five-way fine-grained classification accuracy, indicating that standard VQA benchmark aggregates do not adequately measure this vision-centric capability.

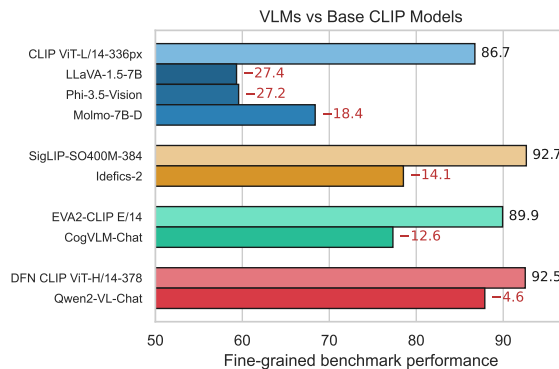


Figure 3: **VLMs compared with their corresponding CLIP vision encoders on fine-grained classification.** Qwen2-VL-Chat nearly matches its DFN-CLIP encoder, whereas the other evaluated VLMs remain well below their associated encoders, revealing substantial fine-grained recognition capacity that is not fully exposed by current multimodal integration.

For each model, we use the default prompt format supplied by VLMEvalKit (Duan et al., 2024) and compute accuracy by exact match against the multiple-choice options. As a vision-only reference, we evaluate CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) using the standard zero-shot classification protocol: images and class labels written as “A photo of a [class]” are encoded by the corresponding image and text encoders, and the top-1 prediction is selected by cosine similarity between the image and text embeddings.

To assess whether existing VLM benchmarks already reflect fine-grained recognition, we compare each model’s average accuracy across the four fine-grained benchmarks with its average score on eight general VQA benchmarks from VLMEvalKit, including MMMU (Yue et al., 2024) and Math-Vista (Lu et al., 2024).

3.4 RESULTS

Figure 2 relates fine-grained classification performance, averaged across the four converted benchmarks, to general VQA performance, averaged across eight existing VQA benchmarks, for every tested VLM. The relationship is weak enough that models with nearly identical general VQA scores can have sharply different fine-grained accuracy. For example, CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) both obtain nearly 48% average accuracy on the general VQA benchmarks, yet their fine-grained classification accuracies differ by 19pp: 77.3% for CogVLM-Chat and 58.1% for LLaVA-NeXT-Vicuna-13B. This discrepancy shows that fine-grained classification is not adequately summarized by standard VLM benchmark aggregates; it is a distinct dimension of visual capability.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Because current VLMs are built on pretrained vision encoders, the multiple-choice protocol also permits a comparison between a VLM and the classification performance of its corresponding CLIP-style encoder. Figure 3 shows that most evaluated VLMs do not preserve the full fine-grained classification ability of their visual backbones under this answer-generation protocol. For example, Molmo (Deitke et al., 2024), despite strong performance on several VQA benchmarks, obtains 68.4% average accuracy in our multiple-choice evaluation, whereas its CLIP ViT-L/14 encoder (Radford et al., 2021) obtains 86.7%. Even Qwen2-VL-Chat (Wang et al., 2024b), the

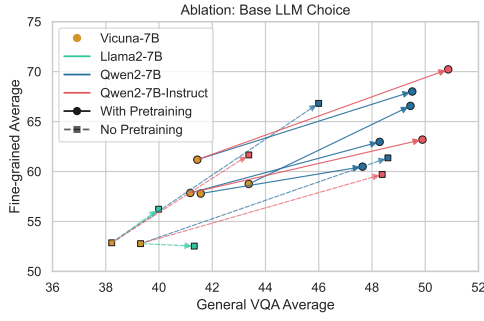


Figure 4: **Base LLM ablation.** Replacing Vicuna-7B in the LLaVA architecture with stronger LLMs improves both evaluation axes. On average, switching from Vicuna to Qwen2-7B increases fine-grained performance by +7.5pp and general VQA performance by +7.7pp.

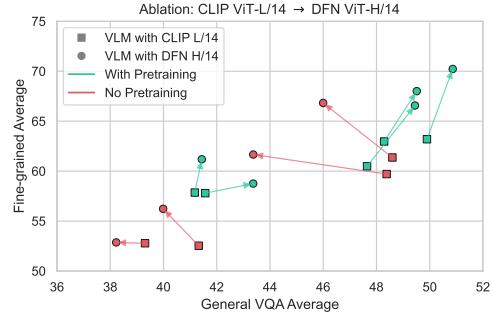


Figure 5: **Vision encoder ablation.** Replacing CLIP ViT-L/14 with DFN-CLIP ViT-H/14 improves fine-grained performance by +4.5pp and general performance by +1.2pp when the encoder is integrated through sufficient pretraining.

strongest evaluated model, remains 4.6 points below its state-of-the-art DFN-CLIP (Fang et al., 2023) vision encoder. This systematic gap motivates the ablation study that follows: improving fine-grained recognition in VLMs requires identifying which model components and training choices determine whether visual-encoder knowledge is retained and made usable by the language model.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

The observational results in Section 3 show that fine-grained classification is not explained by general VQA performance alone. We therefore ablate the model and training factors most likely to account for the observed differences, with emphasis on base model selection and the multimodal training procedure. All experiments use the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023). Unless otherwise specified, models undergo one epoch of pretraining followed by one epoch of finetuning, using the default LLaVA-1.5 hyperparameters for learning rate and batch size.

We report fine-grained performance as the average score across the four classification benchmarks introduced in Section 3.2. To compare these effects with standard multimodal evaluation, we also report the average score on three widely used multiple-choice VQA benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b). These benchmarks provide broad domain coverage and are widely adopted in VLM evaluation, while remaining compatible with the multiple-choice setting used in our fine-grained protocol.

4.2 BASE MODELS

Section 3 suggests that contemporary VLMs increasingly adopt newer base LLMs, while many still rely on the CLIP ViT-L/14 vision encoder introduced by Radford et al. (2021). We isolate how these base-model choices affect fine-grained classification by varying the base LLM and the vision encoder while holding the surrounding LLaVA-style training pipeline fixed.

4.2.1 LLM CHOICE

Setup. Starting from LLaVA-1.5-7B (Liu et al., 2023), we ablate the LLM component across vision encoders and pretraining configurations. The comparison includes Vicuna-7B-1.5 (Chiang et al.,

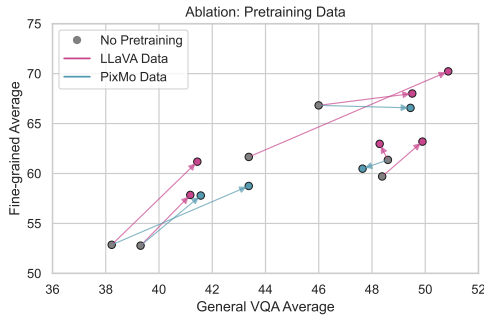


Figure 6: Pretraining data ablation. Adding either LLaVA (CC-3M) or Molmo (PixMo-Cap) pretraining data produces similar gains. LLaVA data increases fine-grained and general scores by +4.0pp and +2.1pp, respectively, whereas PixMo data increases them by +2.4pp and +2.5pp.

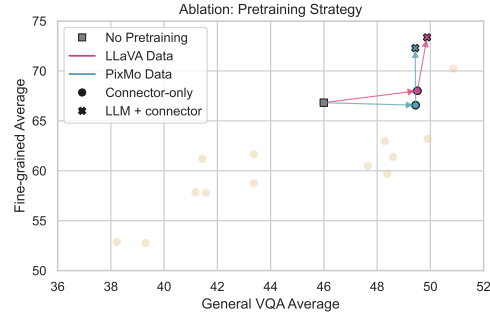


Figure 7: Pretraining update ablation. Compared with connector-only pretraining, unfreezing the LLM after a connector warmup improves fine-grained performance by +5.5pp without reducing general VQA performance.

2023), the original LLaVA LLM; Qwen2-7B (Yang et al., 2024), which is used in Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024); Llama-2-7B (Touvron et al., 2023), the base model from which Vicuna is derived; and Qwen2-7B-Instruct, the instruction-tuned variant of Qwen2-7B.

Findings. Figure 4 shows that replacing Vicuna-7B with Qwen2-7B improves both fine-grained classification and general VQA across configurations, with average gains of +7.5pp on fine-grained benchmarks and +7.4pp on general VQA benchmarks. Qwen2-7B-Instruct produces a similar pattern, improving fine-grained and general VQA performance by +7.5pp and +8.1pp, respectively. Llama2-7B, the non-instruction-tuned base model for Vicuna, yields more modest gains, and the downstream implications of using a non-instruction-tuned model remain less clear. Because all trained models, including the weaker Vicuna-based systems, consistently produce valid multiple-choice outputs, these improvements are best interpreted as gains in the model’s ability to select the correct semantic option rather than gains in response formatting.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. Most evaluated VLMs, including the LLaVA family and Molmo, use OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as the vision encoder. Qwen2-VL (Wang et al., 2024b), in contrast, uses DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which has stronger zero-shot object-recognition performance on benchmarks such as ImageNet. We compare these two encoders across LLM choices and pretraining configurations to determine whether a stronger visual backbone translates into fine-grained gains after multimodal alignment.

Findings. Figure 5 shows that the effect of replacing CLIP L/14 with DFN-CLIP H/14 depends critically on whether connector pretraining precedes finetuning. Without pretraining, DFN-CLIP modestly improves fine-grained benchmark scores (+2.8pp) but reduces general VQA performance (-2.5pp). After connector pretraining, however, general VQA performance is stable or slightly higher (+1.2pp), while fine-grained classification increases substantially (+4.5pp). Thus, stronger vision encoders provide targeted fine-grained benefits, but those benefits require an alignment stage that makes the higher-fidelity visual representation usable by the LLM.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pre-training before the finetuning stage.

Table 1: **Summary of key ablation settings and comparisons with existing models (bolded).** The ablations show that upgrading the base LLM and vision encoder, and pretraining the connector together with the LLM when sufficient data are available, produce the largest gains in fine-grained classification. Instruction finetuning has a smaller effect, whereas general VQA performance is driven more strongly by the choice of LLM.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

4.3 TRAINING METHODS

Modern VLMs are commonly trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). Pretraining uses large-scale image-caption data to align the vision encoder with the LLM through a captioning objective, and instruction tuning subsequently adapts the model with curated multi-round multimodal instruction-completion data. We evaluate how these stages affect fine-grained classification, separating the presence of pretraining, the data source used for pretraining, the parameter set updated during pretraining, and the final instruction-tuning data.

4.3.1 PRETRAINING

Setup. The pretraining ablation addresses three questions: whether LLaVA-architecture models still benefit from image-caption pretraining when their base encoders and LLMs are already pretrained on large-scale data; whether connector-only pretraining is sufficient relative to also updating the LLM; and how much pretraining caption quality affects downstream performance.

We compare three pretraining data configurations: LLaVA pretraining data (Liu et al., 2023), a CC-3M subset with web-scraped captions; Molmo pretraining data (Deitke et al., 2024), specifically PixMo-Cap with detailed human-annotated captions; and no pretraining. In the main pretraining setting, only the connector is updated, following Liu et al. (2023). We also evaluate a strategy similar to Deitke et al. (2024), in which the connector is trained alone for the first 20% of steps and the LLM is unfrozen for the remaining 80%. Every configuration is then finetuned for one epoch on LLaVA finetuning data.

Findings. Figure 6 shows that connector-only pretraining generally improves VLM performance, with the Qwen2 and CLIP ViT-L/14 combination as the main exception. The effect is especially large for Vicuna-based VLMs, where pretraining yields an average improvement of +6.1pp on fine-grained classification and +3.1pp on general VQA. These results identify image-caption pretraining as a major source of fine-grained capability even when the unimodal components are already pre-trained.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

Within the connector-only setting, higher-caption quality does not translate into a clear advantage. Relative to LLaVA data, PixMo data changes fine-grained performance by -1.6pp and general VQA performance by +0.4pp. This result indicates that, under connector-only pretraining, the presence of a pretraining stage matters more than the evaluated difference between web-scraped LLaVA captions and detailed PixMo human annotations.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

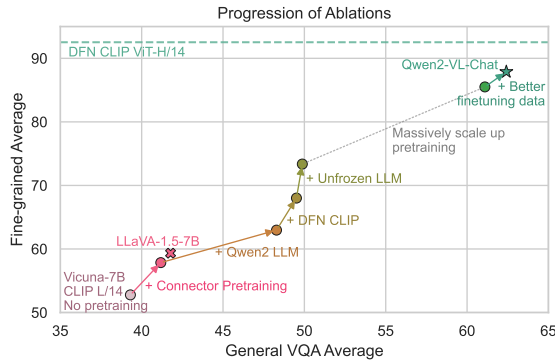


Figure 8: **Progression of combined ablation effects.** Starting from Vicuna-7B and CLIP ViT-L/14, successive changes to the base LLM, vision encoder, and training settings raise fine-grained classification accuracy from 52.8% to 73.4%, closing more than half of the gap to Qwen2-VL-Chat’s 87.9%. The remaining difference is most consistent with Qwen2-VL’s substantially larger pretraining scale of 1.4T tokens.

One explanation is that connector-only pretraining constrains the model’s ability to exploit richer captions because the LLM itself remains fixed. We therefore pretrain both the LLM and connector after a 20% connector-only warmup. As shown in Figure 7, this update strategy improves fine-grained classification by +5.5pp without compromising general VQA scores. The improvement appears with both LLaVA and PixMo pretraining data, indicating that updating the LLM is the decisive factor in this comparison and that training on lower-quality captions does not measurably impair general language-conditioned VQA performance in our setting.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. To isolate the role of instruction-tuning data, we depart from the LLaVA-style base setup and start from Qwen2-VL-7B-Base (Wang et al., 2024b), the pretrained model from which Qwen2-VL-7B-Chat is derived. We finetune this base model on LLaVA-Instruct data (Liu et al., 2023) using the same experimental protocol as above and compare it with Qwen2-VL-7B-Chat, which uses 2M in-house instruction-finetuning examples (Wang et al., 2024b). This comparison holds the pretrained architecture fixed while varying the instruction-tuning data source.

Findings. Finetuning Qwen2-VL-7B-Base only on LLaVA finetuning data slightly reduces both evaluation axes relative to Qwen2-VL-7B-Chat. Fine-grained accuracy decreases by 2.4pp, from 87.9% to 85.5%, and general benchmark performance decreases by 1.3pp, from 62.4% to 61.1%. As summarized in Figure 8, this is the smallest fine-grained effect among the ablated factors, indicating that instruction finetuning contributes less to fine-grained knowledge acquisition than base model selection and pretraining strategy in our experiments.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 synthesize how the ablated components close the gap between weaker and stronger VLMs on fine-grained classification and general VQA. Starting from the LLaVA architecture (Liu et al., 2023) with Vicuna-7B (Chiang et al., 2023) and CLIP ViT-L/14 (Radford et al., 2021), successive modifications raise fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. The pattern is asymmetric: switching to Qwen2-7B (Yang et al., 2024) accounts for most of the general VQA improvement, whereas switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining disproportionately improve fine-grained classification.

A remaining 12-point fine-grained classification gap is not explained by the ablations in Table 1. Two factors could plausibly contribute: architectural differences and pretraining data scale. Although architecture may matter, the observational results in Section 3.4 and the training ablations in Section 4.3 point most strongly to pretraining scale. Our LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) pretraining runs use comparatively small datasets, each with fewer than 1M images and captions, corresponding to approximately 200M and 400M tokens. By contrast, Wang et al. (2024b) report pretraining Qwen2-VL on 1.4T tokens. This orders-of-magnitude difference is the most plausible explanation for the residual gap and reinforces the central conclusion of the ablations: fine-grained classification in VLMs is strongly shaped by the scale and update strategy of multimodal pretraining.

5 LIMITATIONS & CONCLUSION

The study is bounded by the scale at which controlled training ablations are computationally feasible. Our experiments compare pretraining regimes with fewer than 1M image-caption examples rather than the billion-scale or trillion-token regimes used by some recent VLMs, so the analysis cannot directly measure how the identified factors behave under substantially larger pretraining budgets. In addition, newer pretraining strategies may interact differently with fine-grained visual understanding than the LLaVA-style training pipeline evaluated here. These limitations preserve the central conclusion while defining its scope: within the evaluated architectures, data sources, and training settings, fine-grained classification is strongly shaped by base model selection, vision encoder strength, and multimodal pretraining strategy.

This work establishes fine-grained classification as a crucial and undermeasured dimension of VLM evaluation. By evaluating contemporary VLMs on established fine-grained recognition benchmarks and analyzing 22 controlled ablations, we show that fine-grained visual knowledge is not reliably predicted by general VQA performance alone. The evidence further identifies the factors that most directly improve this capability: stronger LLMs improve both general and fine-grained performance, stronger vision encoders disproportionately benefit fine-grained recognition, and pretraining is most effective when the LLM is updated rather than only the connector. These findings provide a concrete empirical basis for building VLMs that preserve precise visual knowledge while retaining strong general multimodal ability.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: **Detailed model-ablation results.** Fine-grained classification and general VQA averages are reported for each combination of vision encoder, LLM, architecture, pretraining data, finetuning data, and parameter-updating strategy.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMUS	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: **Detailed results of model ablations on general VQA benchmarks.**

A FULL RESULTS

We present the complete ablation results for aggregated performance, general VQA benchmarks, and fine-grained classification in Table 2, Table 3, and Table 4, respectively.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: **Fine-grained classification results for model ablations.** The table reports ImageNet, Flowers, Pets, and Food scores for each ablated model configuration.

B EVALUATION DETAILS

B.1 FINE-GRAINED CLASSIFICATION

As described in Section 3.2, our fine-grained evaluation covers four established classification benchmarks and applies a unified multiple-choice protocol to both VLMs and base vision encoders.

ImageNet-1K (Deng et al., 2009): ImageNet-1K is a foundational image classification dataset organized around the WordNet hierarchy and covering a broad range of supercategories, many of which contain fine-grained subcategories. The ImageNet test set contains 50,000 images spanning 1,000 classes.

Because the ImageNet/WordNet hierarchy includes classes that are visually too distinct to support hard multiple-choice alternatives—for example, only three classes fall under the “person” category—we curate the ImageNet subset used in this evaluation rather than converting the full test set directly. Following this constraint, and to avoid an excessively large evaluation set, we use human-annotated incorrect model predictions from Recht et al. (2019) to select images for which hard negative answer choices can be constructed. Starting from 19,056 annotated examples, we retain 5,457 images covering 928 of the original 1,000 classes. Because this curation leaves the retained classes unevenly balanced, we report mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): Oxford Flowers-102 contains 102 flower species and is characterized by substantial intra-class variation together with high inter-class visual similarity. The Flowers test set contains 6,149 examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): Oxford-IIIT Pet-37 contains 37 pet categories and presents challenging variation in pose, lighting, background, and occlusion. The Pets test set contains 3,669 examples.

Food-101 (Bossard et al., 2014): Food-101 contains 101 food categories and exhibits substantial visual diversity within individual classes. The Food test set contains 25,250 examples.

Question generation. For all fine-grained classification datasets except the curated ImageNet subset described above, we generate hard negatives by applying zero-shot classification to the test examples with an OpenCLIP ViT-L/14 model trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). For each example, we select the top four incorrect predictions and shuffle them

with the ground-truth label to form the five answer choices. This construction retains the multiple-choice format required by VLM evaluation while making the alternatives visually and semantically competitive.

Prompt formatting. In the multiple-choice VQA formulation, the base question is “What type of object is in this photo?” Answer choices are formatted as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. Because VLMs are trained with different multiple-choice conventions, we use each model’s corresponding VLMEvalKit prompt format (Duan et al., 2024); for example, some models receive the instruction “Answer with the option’s letter from the given choices directly.”

For CLIP models, no question prompt is used because evaluation follows the standard zero-shot classification procedure. Class names are instead passed to the text encoder using the template “a photo of a {classname}”.

B.2 GENERAL VQA

For the observational analysis in Section 3, we use the scores reported by the OpenVLM Leaderboard (Duan et al., 2024). The leaderboard average aggregates eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). In this paper, that aggregate is used only for Figure 2, where it provides a broad reference point for general VLM performance.

For subsequent evaluations, including both existing models and the models trained in Section 4, we use a more controlled subset of multiple-choice VLM benchmarks to make comparisons with the fine-grained protocol more direct. The subset consists of the following benchmarks.

MMBench (Liu et al., 2024b): MMBench contains 2,948 examples that evaluate a broad range of VLM capabilities across perception and reasoning. Although most examples concern coarse-grained perception and visual reasoning, the benchmark also includes examples labeled as “fine-grained perception”; these categories, such as action recognition, attribute recognition, and OCR, differ from the object-category classification datasets evaluated in this work.

MMMU (Val) (Yue et al., 2024): MMMU is a large-scale multimodal benchmark spanning a wide distribution of domains and applications. The full test set contains 10,500 examples, while model performance is commonly reported on the validation split of 900 questions. The benchmark evaluates general knowledge tied to visual perception across art, science, engineering, medicine, business, and the humanities.

MMStar (Chen et al., 2024b): MMStar contains 1,500 questions selected by human annotators to cover diverse domains and low-level multimodal capabilities. We include it because it provides a complementary multiple-choice measure of general VLM performance.

Evaluation procedure. For the general VQA benchmarks, we use the VLMEvalKit evaluation code (Duan et al., 2024). Existing models are evaluated with their predefined procedures, and our trained models are evaluated with the LLaVA evaluation code because all of them are finetuned on LLaVA-Instruct data.

C TRAINING

All controlled training experiments build on the LLaVA-1.5 training codebase (Liu et al., 2023). Unless otherwise stated, we retain the LLaVA architecture and training procedure so that individual ablations can be attributed to the intended model or data change rather than to unrelated implementation differences.

Pretraining. Models are pretrained for one epoch on either the LLaVA pretraining data, obtained from CC-3M images and captions, or the Molmo pretraining data, PixMo-Cap. For PixMo-Cap, we use only the “caption” field and exclude the raw audio transcript. Pretraining uses a batch size of 256. In most runs, only the randomly initialized MLP connector is updated, with a learning rate of $1e-3$. In the pretraining-strategy ablation, we first train only the connector for a 20% warmup period,

then unfreeze the LLM and train it together with the connector for the remaining 80% of steps using a learning rate of $2e-5$ for the LLM.

Finetuning. All models are finetuned for one epoch on the LLaVA-1.5 finetuning data, which consists of LLaVA-Instruct and VQA training data from multiple datasets used to support benchmarking. Finetuning uses a batch size of 128 and a learning rate of $2e-5$ for the LLM. The vision encoder remains frozen during both pretraining and finetuning.

For the Qwen2-VL architecture ablation, we initialize from Qwen2-VL-7B Base and apply the same finetuning procedure, unfreezing only the LLM.

Compute. All models are trained on 4 nodes with 4 A100 (40GB) GPUs per node. Training typically completes within 20 hours, with variation determined primarily by the size of the CLIP model.

LLaVA vs. PixMo data. The main text contrasts the caption quality of LLaVA and PixMo pretraining data. LLaVA pretraining data is drawn from CC-3M and consists of web-scraped captions, which are commonly noisy and only loosely aligned with image content. PixMo-Cap, by contrast, is constructed from human annotators recording a 60–90 second audio description of an image, which is then converted into an image caption (Deitke et al., 2024). A surface comparison illustrates the difference in density: LLaVA captions average 9.8 words, whereas PixMo captions average 169 words.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”